

OVIDIUS UNIVERSITY OF CONSTANȚA
Faculty of Mathematics and Computer Science · Computer Science

BACHELOR'S THESIS

Information System for Network Detection Intrusion

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Project purpose

❝ Real-time detection

Each packet inspected as it arrives. No PCAP files analysed offline.

★ Hybrid signatures + ML

Two complementary engines. When both agree, the alert gets the hybrid label.

☆ Built-in forensics

Per alert: a snapshot of the last 50 packets, downloadable from the UI.

➥ Easy demo

One-click launcher: auto-install, FastAPI + Vite start, browser opens.

02 · IDEA

Project idea

Neither classical paradigm solves the problem on its own.

<>

SIGNATURES

Precise but limited

Catch exactly what they know (CVEs, Snort signatures). Cannot see new or modified attacks.

★

MACHINE LEARNING

Flexible but noisy

Detects unknown anomalies. But generates false alarms — see Sommer & Paxson, 2010.

✔

HYBRID

Explicit cooperation

When both engines flag the same packet → the hybrid category (maximum confidence).

Plus a built-in forensics layer — every alert keeps

Technologies used

Backend

Python 3.12	Main language, asyn
FastAPI+Uvicorn	REST + WebSocket
SQLAlchemy 2	Async ORM + aiosq
scikit-learn	Random + Isolation
Scapy	Real-time packet cap
bcrypt + JWT	Auth + password has

Frontend

React 18 + Vite	SPA, hot reload
TailwindCSS	Deep Navy + Cyan theme
Recharts	Live charts
lucide-react	Iconography
WebSocket client	Streaming alerts/traffic
date-fns	Timestamp formatting

04 · ARCHITECTURE

Packet flow through the system



Capture

Two interchangeable sources: LiveSniffer (Scapy) or TrafficSimulator. Same API.

Processing

Dispatcher: metrics, 200-packet ring buffer, traffic_bus publish, DB sampling.

Detection

Signature engine (10 YAML rules) + ML scoring. Three categories.

Persistence

Alerts in SQLite + JSONL per alert on disk. Broadcast on alert_bus.

05 · COMPONENTS

The modules that make up the application

❝ Sniffer / Simulator

Real packet capture (Scapy) or synthetic generation for demos.

■ Dispatcher

The central point. Coordinates metrics, engines, persistence, broadcast.

✔ Signature Engine

Editable rules: protocol, IP, port, content, regex, threshold.

★ ML Detector

Random Forest or Isolation Forest on 6 features per packet.

☆ Forensics

200-packet ring buffer; per alert, JSONL written to disk.

❍ API + WebSocket

JWT-protected REST + /ws/alerts + /ws/traffic streaming.

The detection pipeline

Per-packet processing

1. `Dispatcher.handle(pkt)` on every captured packet.
2. Increment counters + update PPS (1 s sliding window).
3. Packet enters the 200-slot ring buffer.
4. Publish on `traffic_bus` → live dashboard chart.
5. 1/5 sampling persisted as TrafficEvent

Alert categories

SIGNATURE

Only the signature engine fired. Precise, auditable.

ML

ML scored $\geq$ 0.8. Suspicious packet, no known rule.

HYBRID

Two complementary algorithms

✔ Random Forest

supervised

LEARNING	Labelled data (attack vs benign)
OUTPUT	Label + calibrated probability
STRENGTH	Interpretable feature importance
WEAKNESS	Cannot detect unknown attacks

★ Isolation Forest

unsupervised

LEARNING	Normal traffic only — no attack labels
OUTPUT	Anomaly score (higher = suspicious)
STRENGTH	Detects unknown outliers
WEAKNESS	Higher false-positive rate

Per-alert capture

How it works

1. Ring buffer

The dispatcher keeps a FIFO of the last 200 captured packets.

2. On every alert

The last 50 packets are extracted and serialised as JSON Lines.

3. Association

The file is linked to the alert via a foreign key.

4. Download

Authenticated endpoint

`alert_12_1716387210.jsonl`

```
{"ts": "...:32.481Z", "protocol": "TCP",
 "src_ip": "45.77.12.88", "src_port":
51232,
 "dst_ip": "10.0.0.10", "dst_port": 22,
 "length": 60, "flags": "S",
 "payload_preview": ""}
{"ts": "...:32.491Z", "protocol": "TCP",
 "src_ip": "45.77.12.88", "src_port":
51233,
 "dst_ip": "10.0.0.10", "dst_port": 80,
 "length": 60, "flags": "S",
 "payload_preview": ""}
...
// 50 packets total
```

09 · USAGE EXAMPLE

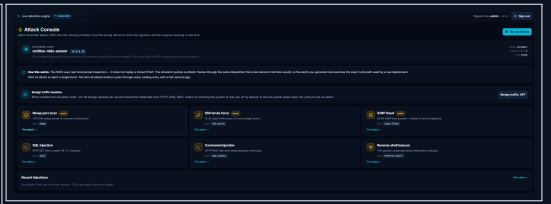
Dashboard and Attack Console

DASHBOARD



Live KPIs, packet-rate chart, severity pie, alert timeline, live feed. Monitoring host `10.0.0.10`.

ATTACK CONSOLE



Six attack buttons. Defended host + benign-traffic toggle. Recent injections audit trail.

10 · ALERT FLOW

From attack to triage

1 · Attack triggered

User clicks “Nmap port scan” in the Attack Console. The simulator generates 30 TCP SYN packets from `185.220.101.5` toward `10.0.0.10`.

2 · Capture

The dispatcher receives each packet, updates the ring buffer and publishes on `traffic_bus` for the live chart.

3 · Signature detection

After the 8th packet, the “Nmap TCP SYN scan” rule crosses the threshold (8 SYNs in 5 s) — the alert is created.

4 · Persistence + forensics

The alert is saved in SQLite. JSONL ring-tail with the last 50 packets written to disk and linked to the alert.

6 · Triage

Similar applications on the market

Feature	Snort	Suricata	Zeek	Wazuh	Splunk	Ovidius NIDS
Signature engine	✔	✔	~	~	~	✔
Machine Learning engine	✗	✗	✗	~	✔	✔
Hybrid category	–	–	–	✗	✗	✔
Built-in analyst UI	✗	✗	✗	✔	✔	✔
Per-alert forensic capture	✗	✗	✗	✗	✗	✔
Real-time streaming in UI	✗	✗	✗	~	✔	✔
Rules editable from UI	✗	✗	✗	✔	✔	✔
One-click deploy	✗	✗	✗	✗	✗	✔
Open source · free	✔	✔	✔	✔	✗	✔

All in one binary *— detection, ML, forensics and UI;* ***forensics built-in*** *per alert;* ***explicit hybrid*** *category;* ***five-minute demo*** *via one-click launcher.*

What was achieved

✔ End-to-end functional

Capture, hybrid detection, forensics, auth, live dashboard — all operational.

■ Real-time, low latency

< 1 ms per packet at signature engine, < 250 ms total to UI.

★ ML trainable from UI

Random Forest and Isolation Forest, metrics and feature importance.

☆ Forensics on disk

Per alert: JSONL with the last 50 packets, authenticated download from UI.

➥ One-click deployment

`run.bat` / `run.sh` / `run.py` — auto install + start.

<> Extensible code

Rules editable without restart. Decoupled modules. Auto OpenAPI.

What is left for future development

PRIORITY

Per-flow features

Switch from per-packet to per-flow 5-tuple. Improves multi-packet attack precision.

STRETCH

Alert correlation

Group consecutive alerts from the same source into a single “incident”.

PRIORITY

Evaluation on CICIDS2017

STRETCH

Snort rule import



Thank you for your attention!

Bora Cristian-Vasile · Universitatea OVIDIUS din Constanța · Iulie 2026